

Naive Method

Recurrence: None

Time complexity: $\theta(n^3)$

Divide and conquer

Recurrence: $T(n) = 8T(n/2) + \theta(n^2)$ if $n > 1$. $T(n) = \theta(1)$ if $n=1$

Time complexity: $T(n) = \theta(n^3)$

Strassen Algorithm

Recurrence: $T(n) = 7T(n/2) + \theta(n^2)$ if $n > 1$. $T(n) = \theta(1)$ if $n=1$

Time Complexity: $T(n) = \theta(n^{\log_2(7)})$